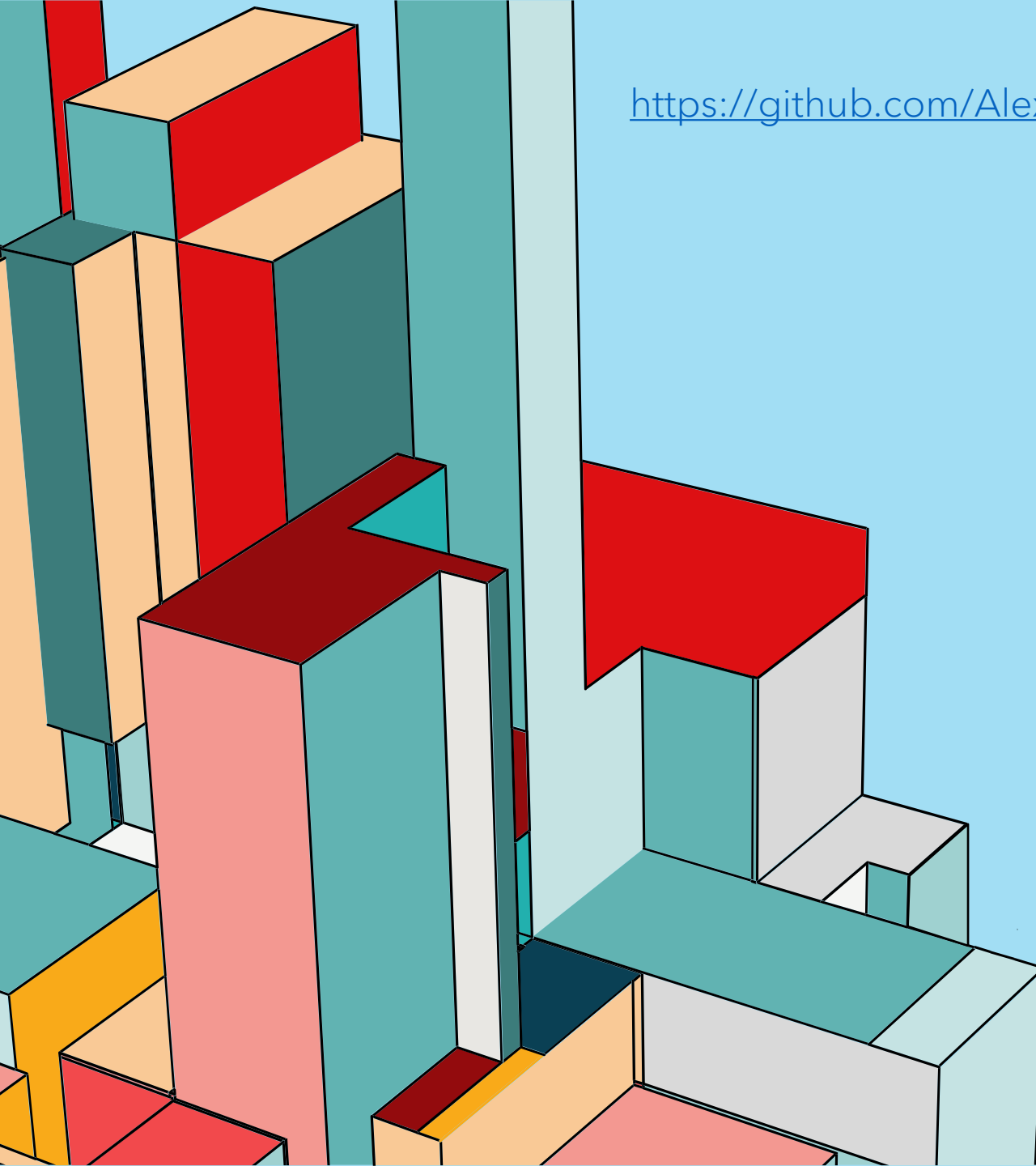




https://github.com/AlexPoloAI/Alex_Polozov_ML_ITAI1371_12321_FINAL

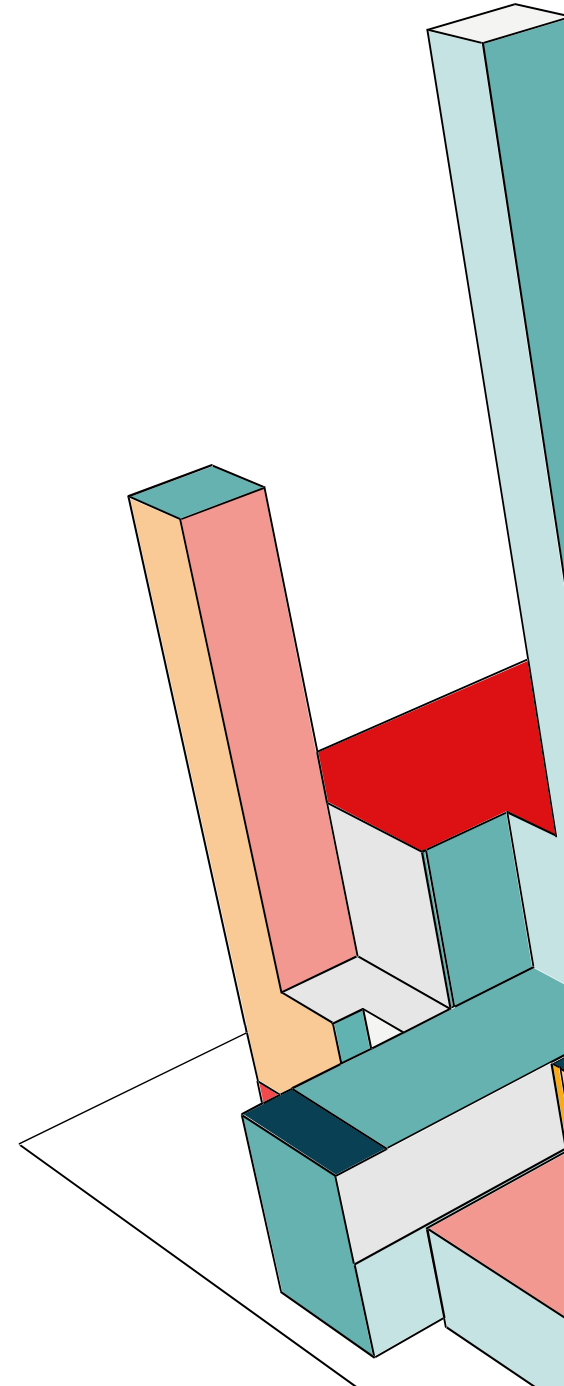
FINAL EXAM

ALEX POLOZOV
PROFESSOR VISWANATHA RAO (VISHWA)
6263-ITAI-1371-INTRO TO MACHINE
LEARNING-S10-12321
AUGUST 7, 2026

APPROACH & DATA

- I plan to predict house prices.
- This task is one regression problem.
- I cleaned this data and filled empty cells.
- I divided this data into Train, Validation, and Test parts

70%/15%/15%



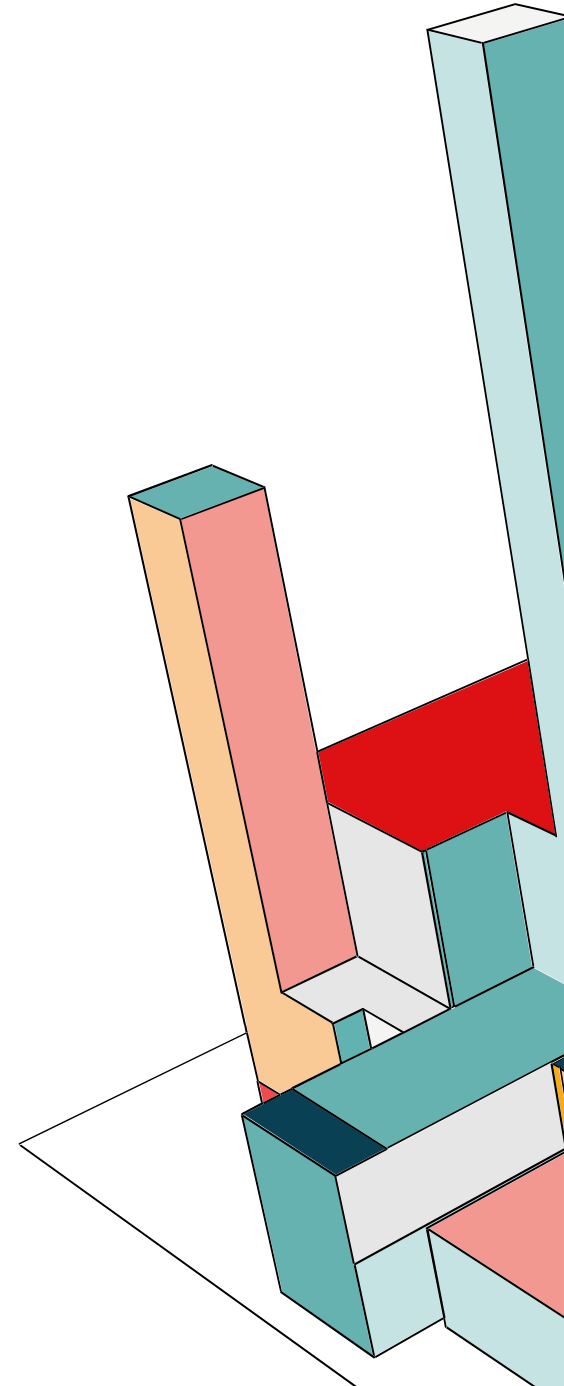
BASE MODELS & ENSEMBLE MODELS

I used Train data to teach five base models. I checked every model on Validation data and Test data:

- Gradient Boosting showed this best result. It scored 0.8809 R2.
- K-Nearest Neighbors showed this worst result. It struggled to measure distance

I tried to build two ensemble models:

- Voting Regressor combined three good models to score 0.8513 R2.
- Bayesian Ridge scored 0.8789 R2. It uses math probability to guess prices well.



FINAL CHOICE MODEL

- Gradient Boosting won this competition.
- It achieved highest test score. It produced lowest error.
- I plan to use this model to predict any new house prices.

Model	MAE	MSE	Rsquared (R2)
Gradient Boosting Regressor	0.0984	0.0208	0.8809
Bayesian Ridge (Ensemble)	0.1018	0.0211	0.8789
Random Forest Regressor	0.1052	0.0248	0.8578
Voting Regressor (Top 3)	0.1093	0.0259	0.8513
Decision Tree Regressor	0.1593	0.0578	0.6694
Linear Regression	0.1808	0.0612	0.6499
K-Nearest Neighbors Regressor	0.1874	0.0675	0.6137

Thank you FOR THIS CLASS

